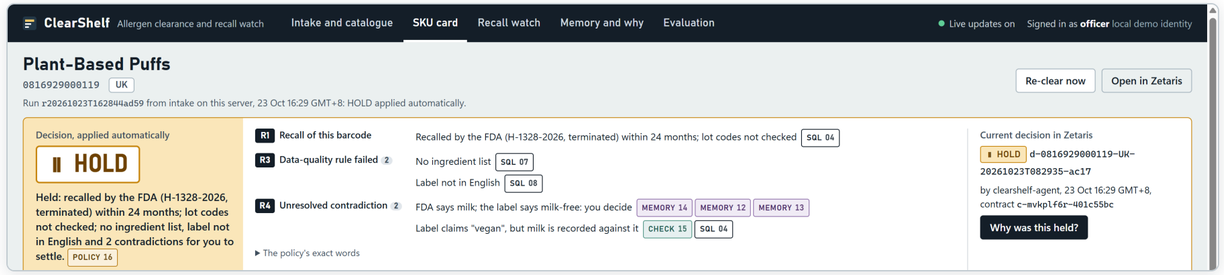
ClearShelf

Allergen clearance and recall watch for online grocers. Every decision is cited and every release is signed.

Zetaris	The data harness: Open Food Facts, openFDA and UK FSA read only through Lightning SQL; the clearance USL's data-quality rules gate every clearance.
Meterless	Markovian investigation loop, H-MEM memory and ledger, World Model timeline, Scout-style signed execution contracts.
NVIDIA	Nemotron 3 Nano 4B on a local GPU reads labels and notices and plans each investigation. 0 cloud model calls.



The SKU card for Plant-Based Puffs (UK): held automatically on an FDA milk recall, a missing ingredient list and a vegan claim the recall contradicts. Local ClearShelf UI, 23 Oct 2026.

Umar Mohktar, Singapore – solo

Data: Open Food Facts (ODbL 1.0), openFDA (public domain / CC0, not endorsed by the FDA), UK FSA (Open Government Licence v3.0).

One officer, hundreds of listings, sources that disagree

The user: a product-compliance officer at a UK online grocer

The grocer also sells US- and EU-made packaged food. The officer signs off the allergen information shown before purchase, which Reg. (EU) 1169/2011 Art. 14 requires for distance selling (retained in UK law), and pulls or relabels live listings when an FSA or FDA notice lands.

The problem: label data and recall notices disagree, and the check is manual

Open Food Facts lists **Plant-Based Puffs** (UPC 816929000119) as vegan, with no allergen tags and no ingredient list. FDA recall **H-1328-2026** (Class I, reported 2026-09-09) says the product **"tested positive for undeclared milk"**. Reconciled by hand, a wrong "free-from" listing can stay live.

6,325

openFDA food enforcement reports whose recall reason matches "undeclared"

842 of the newest 1,000 concern an allergen; counted 23 Oct 2026

~7 a month

UK FSA allergy alerts: 200 from 8 Jul 2024 to 21 Oct 2026

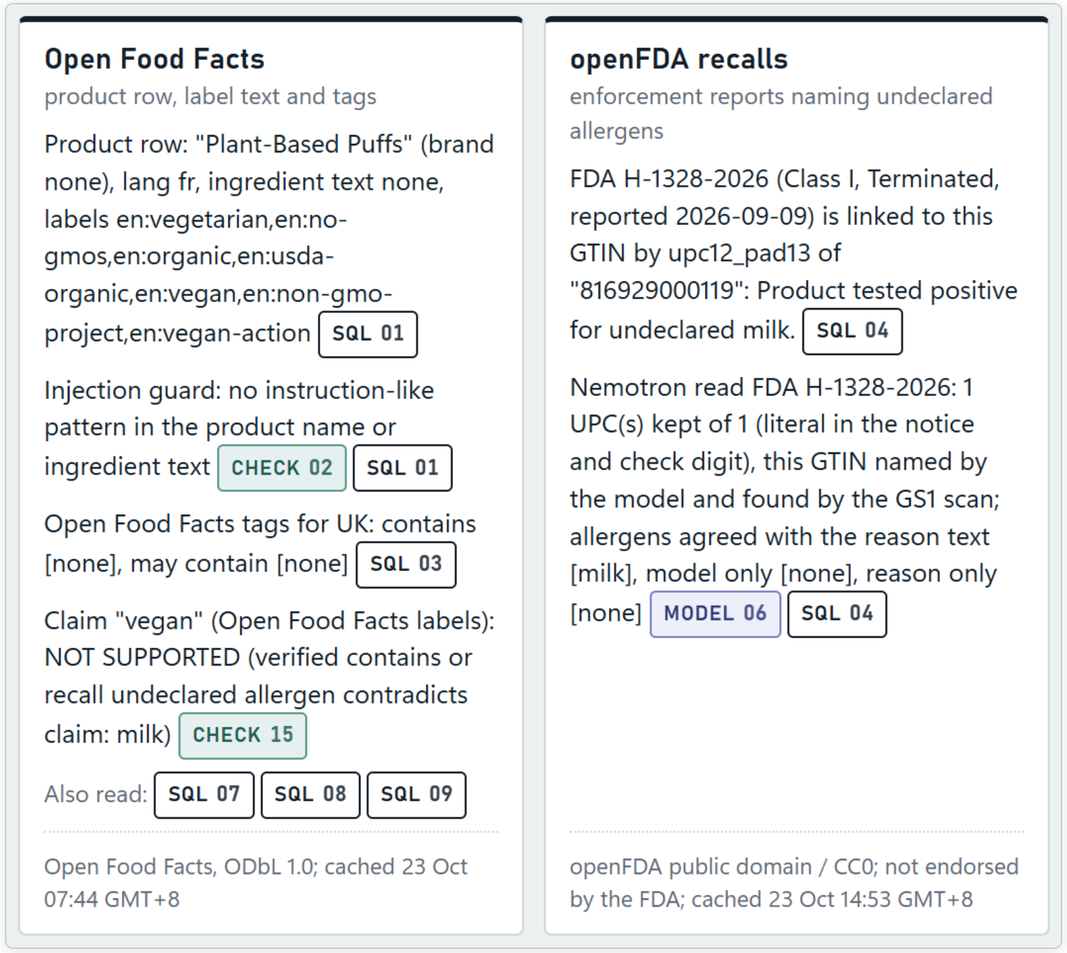
counted in Zetaris and in the source JSON: equal

17 of 17

catalogue listings whose barcode an FDA recall names, held or blocked automatically

the 30-product catalogue includes 14 barcodes taken from FDA recalls; none proposed for clearance

ClearShelf reconciles these sources for each product and market, cites every claim, holds anything unsettled on its own, and clears only with the officer's signature.



Both sources on one card: the Open Food Facts row (vegan labels, no ingredient text) and the openFDA recall (undeclared milk). Every line carries its evidence chip.

Counts measured on 23 Oct 2026 through Zetaris; see zetaris/README.md. Open Food Facts data, ODbL 1.0.

WHY THIS NEEDS AN AGENT

Reading, contradictions, memory and risk need an agent

1 Read, don't keyword-match

Cocoa butter is not milk. "May contain" is precautionary. Labels arrive in German, French and Italian.

2 Contradictions go on the record

Source against source, model against verifier, memory against source, rule against rule: each is queued for the officer, never settled by the last write.

3 Memory carries rulings forward

Officer rulings and verified extractions persist in H-MEM. New standing rules come only from proposals a human approves.

4 Actions are asymmetric

The agent may HOLD or BLOCK on its own, the safe direction. CLEAR and RELEASE need the officer's signed contract.

The label as read ingredients_text from Open Food Facts SQL 01

Ingredients

... cocoa butter✓ K ...

ALLERGY ADVICE: May contain

peanuts✓ K, milk✓ K and nuts✓ K

contains✓ may contain✓ not an allergen✓ K keyword rule hit ✓ model and verifier agree, ✗ they disagree (held for the officer)

Nemotron named 3 spans; the verifier agreed on 3 of 3 MODEL 04 CHECK 05

> **Keyword baseline**, for comparison only (it decides nothing): it would mark **milk, peanuts, tree nuts** as contains, from the words marked K above.
>
> - "butter" is inside "cocoa butter": the keyword rule calls it milk; the verifier reads it as not an allergen (cocoa butter is not milk fat).
> - "peanuts": the keyword rule says contains; the label says may contain (precautionary).
> - "milk": the keyword rule says contains; the label says may contain (precautionary).
> - "nuts": the keyword rule says contains; the label says may contain (precautionary).

Waitrose Dark Chocolate, UK. The keyword rule (K) calls cocoa butter milk and reads "may contain" as contains; Nemotron and the lexicon verifier agree on three precautionary spans. Label text from Open Food Facts, ODbL 1.0, shortened here to a quote (...).

5/15 → 0/15

trap false alarms, keyword baseline against Nemotron

0/27 → 18/27

"may contain" read as exactly may-contain

Measured on 27 SKUs, 23 Oct 2026 (docs/eval-report.md, section 3).

A single prompt cannot query anything; a script cannot weigh a crowd-sourced vegan claim against a lab-tested recall and an earlier ruling. Trap words scored: cocoa butter, cream of tartar, lactic acid, plant milks, coconut, nutmeg, shea, buckwheat, maltodextrin.

SOLUTION WALKTHROUGH

From barcode to a cited decision, then a watch on every new notice

Intake
Barcode (EAN-13, UPC-A, EAN-8 or GTIN-14) and market (UK or US). The GS1 check digit is verified first.

›

Register in Zetaris
The product's Open Food Facts table is created, described, cached and added to the USL.

›

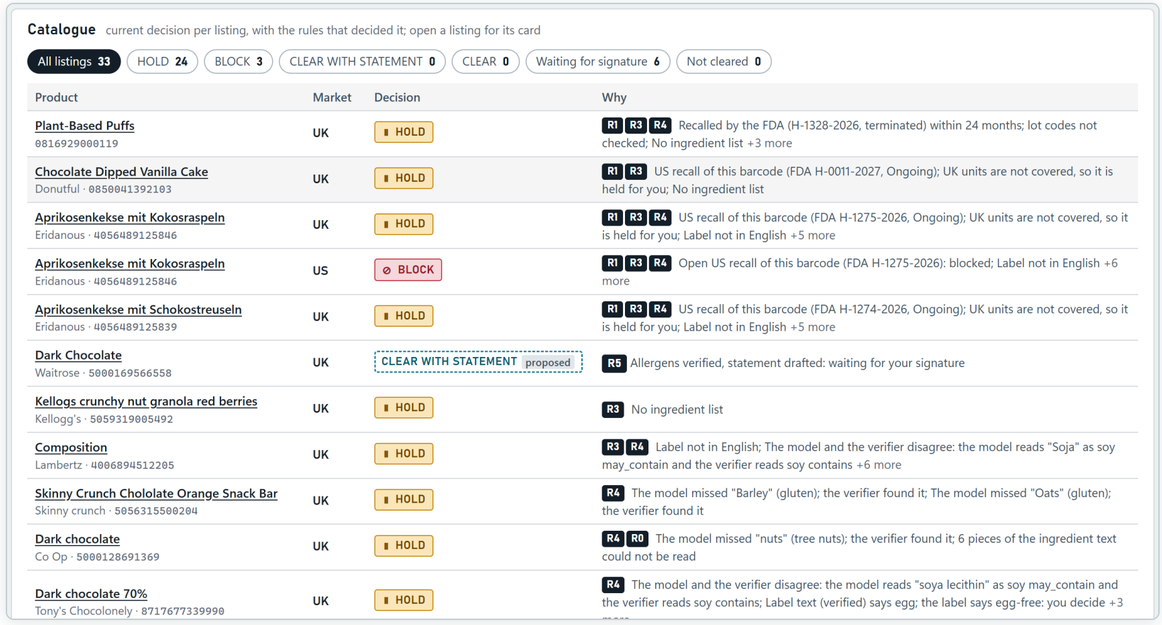
Investigate
Markovian loop on Nemotron: product, guard, label, verifier, recalls, notices, DQ gate, memory.

›

Decide
Deterministic policy R0–R6. The model drafts; it never decides.

›

Act and watch
HOLD or BLOCK is applied and written back; a clearance waits for a signature. Each new notice is matched against the catalogue.

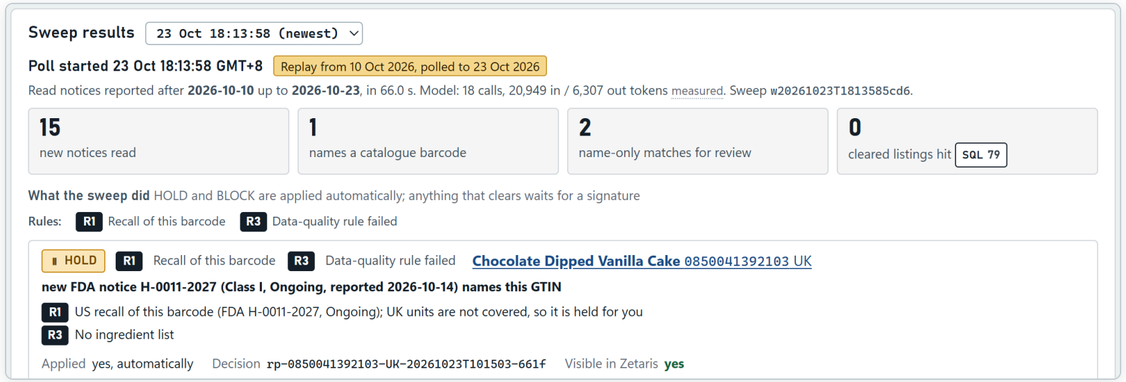


Catalogue current decision per listing, with the rules that decided it; open a listing for its card

All listings 33 | HOLD 24 | BLOCK 3 | CLEAR WITH STATEMENT 0 | CLEAR 0 | Waiting for signature 6 | Not cleared 0

Product	Market	Decision	Why
Plant-Based Puffs 0816929000119	UK	HOLD	R1 R3 R4 Recalled by the FDA (H-1328-2026, terminated) within 24 months; lot codes not checked; No ingredient list +3 more
Chocolate Dipped Vanilla Cake Donutful · 0850041392103	UK	HOLD	R1 R3 US recall of this barcode (FDA H-0011-2027, Ongoing); UK units are not covered, so it is held for you; No ingredient list
Aprikosenkekse mit Kokosraspeln Eridanous · 4056489125846	UK	HOLD	R1 R3 R4 US recall of this barcode (FDA H-1275-2026, Ongoing); UK units are not covered, so it is held for you; Label not in English +5 more
Aprikosenkekse mit Kokosraspeln Eridanous · 4056489125846	US	BLOCK	R1 R3 R4 Open US recall of this barcode (FDA H-1275-2026): blocked; Label not in English +6 more
Aprikosenkekse mit Schokostreuseln Eridanous · 4056489125839	UK	HOLD	R1 R3 R4 US recall of this barcode (FDA H-1274-2026, Ongoing); UK units are not covered, so it is held for you; Label not in English +5 more
Dark Chocolate Waitrose · 5000169566558	UK	CLEAR WITH STATEMENT proposed	R5 Allergens verified, statement drafted: waiting for your signature
Kellogs crunchy nut granola red berries Kellogg's · 5059319005492	UK	HOLD	R3 No ingredient list
Composition Lambertz · 4006894512205	UK	HOLD	R3 R4 Label not in English; The model and the verifier disagree: the model reads "Soja" as soy may_contain and the verifier reads soy contains +6 more
Skinny Crunch Chololate Orange Snack Bar Skinny crunch · 5056315500204	UK	HOLD	R4 The model missed "Barley" (gluten); the verifier found it; The model missed "Oats" (gluten); the verifier found it
Dark chocolate Co Op · 5000128691369	UK	HOLD	R4 R0 The model missed "nuts" (tree nuts); the verifier found it; 6 pieces of the ingredient text could not be read
Dark chocolate 70% Tony's Chocolonely · 8717677339990	UK	HOLD	R4 The model and the verifier disagree: the model reads "soya lecithin" as soy may_contain and the verifier reads soy contains; Label text (verified) says egg; the label says egg-free: you decide +3

The catalogue after intake: 33 listings (30 products, UK and US). 17 cite an exact-barcode recall and were held or blocked; 10 more were held or blocked for other reasons; 6 wait for the officer's signature. Each row shows the rules that decided it.



Sweep results 23 Oct 18:13:58 (newest)

Poll started 23 Oct 18:13:58 GMT+8 Replay from 10 Oct 2026, polled to 23 Oct 2026

Read notices reported after **2026-10-10** up to **2026-10-23**, in 66.0 s. Model: 18 calls, 20,949 in / 6,307 out tokens measured. Sweep w20261023T1813585cd6.

15	1	2	0
new notices read	names a catalogue barcode	name-only matches for review	cleared listings hit SQL 79

What the sweep did HOLD and BLOCK are applied automatically; anything that clears waits for a signature

Rules: R1 Recall of this barcode R3 Data-quality rule failed

HOLD R1 Recall of this barcode R3 Data-quality rule failed **Chocolate Dipped Vanilla Cake** 0850041392103 UK

new FDA notice H-0011-2027 (Class I, Ongoing, reported 2026-10-14) names this GTIN

R1 US recall of this barcode (FDA H-0011-2027, Ongoing); UK units are not covered, so it is held for you

R3 No ingredient list

Applied yes, automatically Decision rp-0850041392103-UK-20261023T101503-661f Visible in Zetaris **yes**

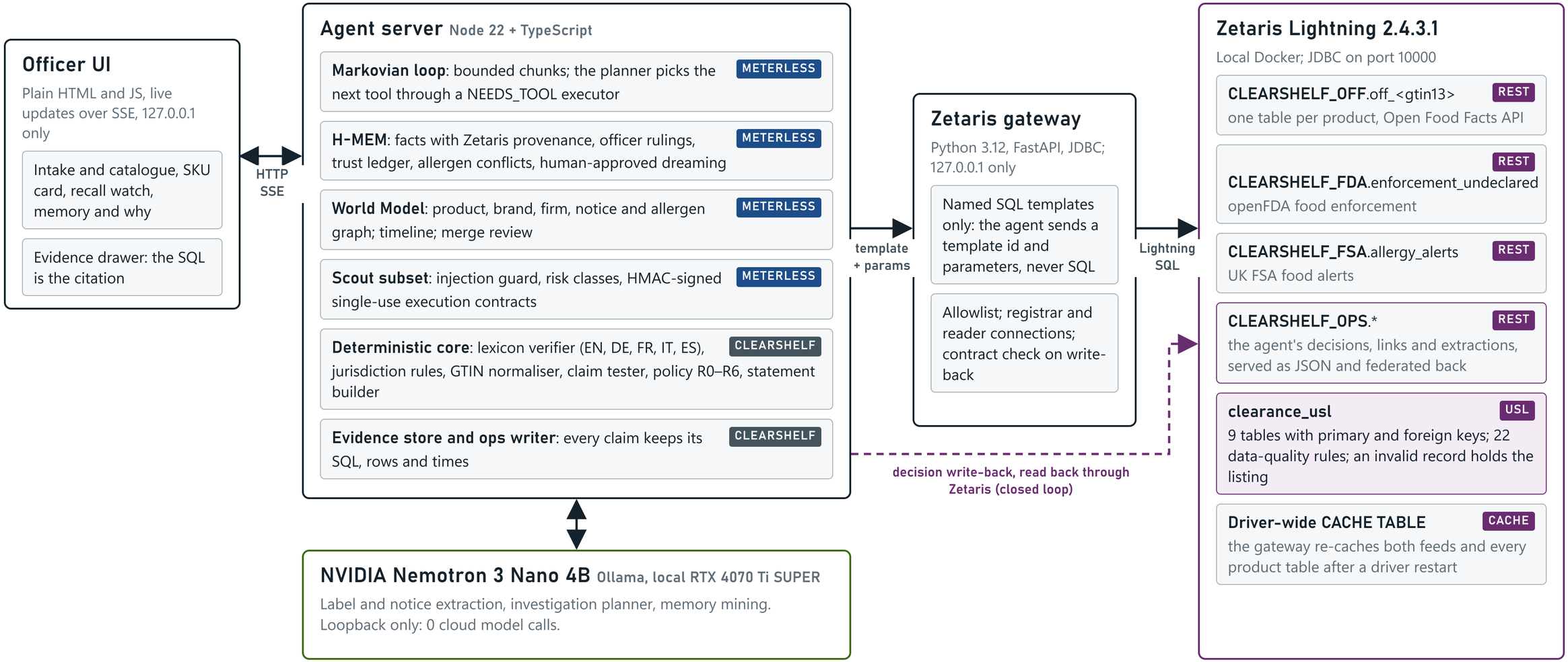
Recall watch, labelled as a replay from 10 Oct polled to 23 Oct: 15 new notices, 1 exact barcode match, held automatically (a US-only recall does not cover UK units, so the UK listing is held for the officer). Notice to hold: median 68.4 s over 10 replayed polls.

BLOCK	automatic
HOLD	automatic
CLEAR WITH STATEMENT	signed
CLEAR	signed

RELEASE of a hold also needs a signed contract. Open Food Facts stands in for the grocer's supplier or PIM feed.

BLOCK: an active recall of this barcode covers this market. HOLD: waits for the officer. R0 investigation incomplete · R1 recall of this barcode · R2 name-only match · R3 data-quality rule failed · R4 unresolved contradiction · R5 verified allergens, clear with statement · R6 no allergens, clear.

Everything runs on one machine; every source read goes through Zetaris



At runtime the agent never calls Open Food Facts, openFDA or the FSA directly; only Zetaris does. Ports: UI 8790, gateway 8791, Ollama 11434, Zetaris JDBC 10000.

Zetaris is the data harness, the gate and the citation

1 Federated REST tables, counted against the source

Four Lightning databases: openFDA (1,000 reports), UK FSA (200 alerts; double explode gives 349 alert-allergen rows), one Open Food Facts table per product (30 catalogue products; 40 registered, including 10 eval-only), and the ops store. Zetaris counts equal the source JSON counts.

2 Agent-driven registration

A new barcode at intake makes the registrar create its Open Food Facts table, DESCRIBE it, CACHE it and re-activate the USL product table. The cache is driver-wide: a cached openFDA count takes about 0.1 s, against 0.9 to 2.8 s uncached.

3 The USL is the contract, its data-quality rules are the gate

`clearance_usl`: 9 tables, 22 DQ rules (9 primary key, 8 foreign key, 5 custom). Any invalid record for the product holds it under R3. At onboarding: has_ingredients 3 of 30 invalid, english_label 5 of 30, market_code 6 of 111.

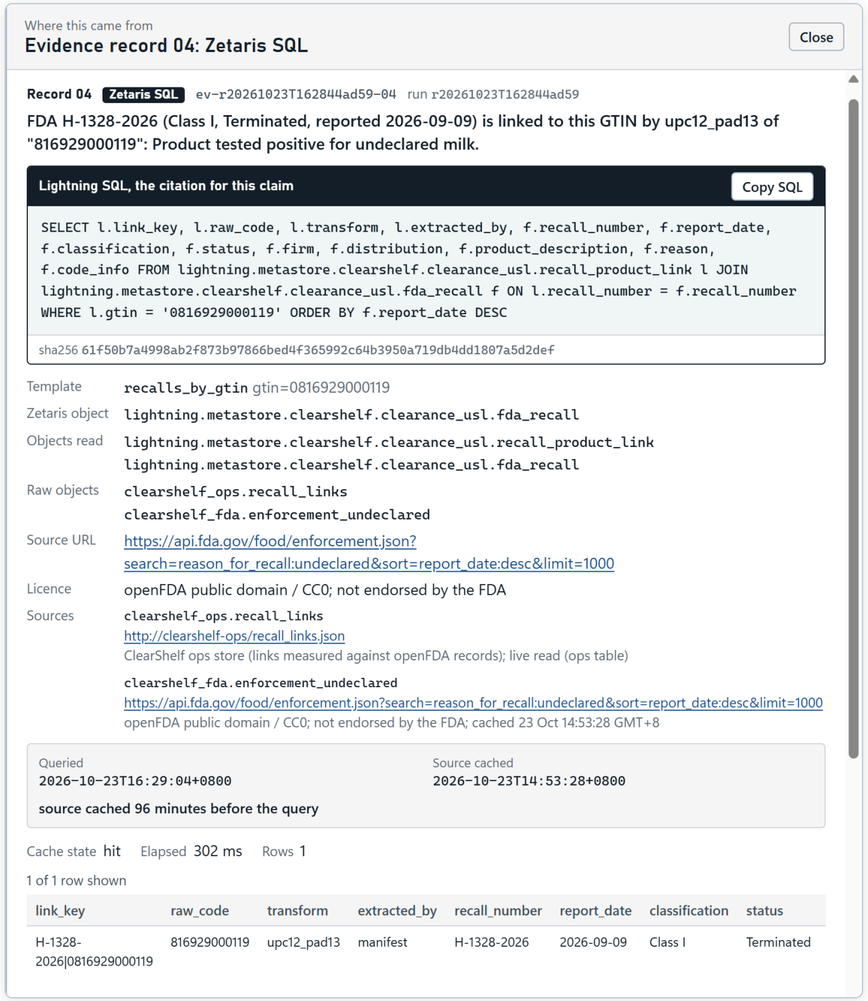
4 The SQL is the citation

Every claim on a card links to the Lightning SQL that produced it, with its sha256, objects, source URL, licence and cache time. Re-running all 95 cited Zetaris and DQ records returned every cited row (95/95).

5 Closed loop, checked by Zetaris

Decisions are federated back as `clearance_usl.decision` and re-read in the "why" trace and the recheck. After every signed write, the `signed_clear` DQ rule checks the row: a signed clearance passed, a synthetic unsigned row was flagged. Join map: 11 UPC-12 and 3 EAN-13 links, 0 orphans.

Limitation: the least-privilege reader role is not configured, so reads run on the admin login behind the gateway's read-only allowlist.



Evidence record 04: the Lightning SQL that links FDA H-1328-2026 to the Puffs barcode, with sha256, objects, source URL, licence, cache time and the returned row.

Counts from zetaris/README.md (full onboarding, 23 Oct 2026) and docs/eval-report.md (E-replay). Rules in zetaris/dq-rules.md.

Meterless engines run the reasoning loop, the memory and the audit trail

Markovian: bounded investigation

Vendored reference, unmodified. Nemotron plans the next tool call; each chunk carries a bounded state (768-token budget, peak 474, estimated). Measured prompts stay between 1,322 and 1,788 tokens per chunk. Two investigations re-sent with their full history measured 4,641 and 6,829 tokens by the last chunk: the Markovian prompts were 48% and 58% smaller in total.

H-MEM: memory, conflicts and the ledger

Vendored reference plus an allergen-conflict module. Facts carry Zetaris provenance; conflicts are always queued (11 pairs now); the ledger answers "Why was this held?". A warm run reused a verified extraction: tool-model tokens 3,658 → 804.

World Model: what we knew when

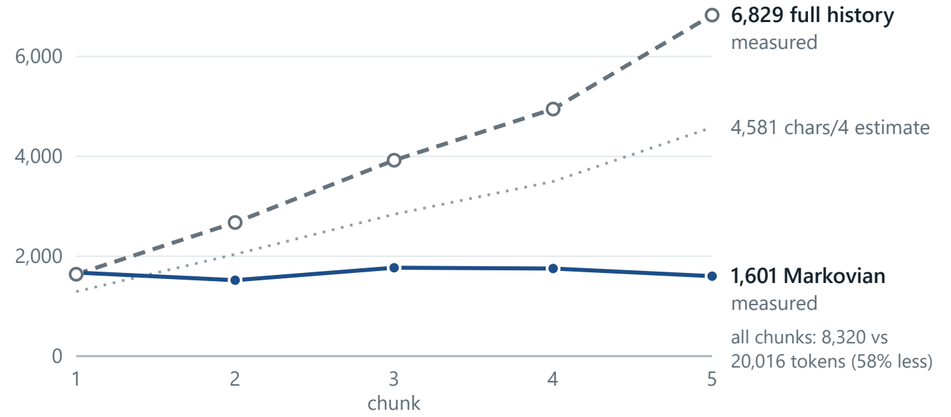
Vendored reference. Product, brand, firm, notice and allergen graph with a timeline and a merge review queue; re-ingesting the catalogue and live Zetaris rows gives an empty diff.

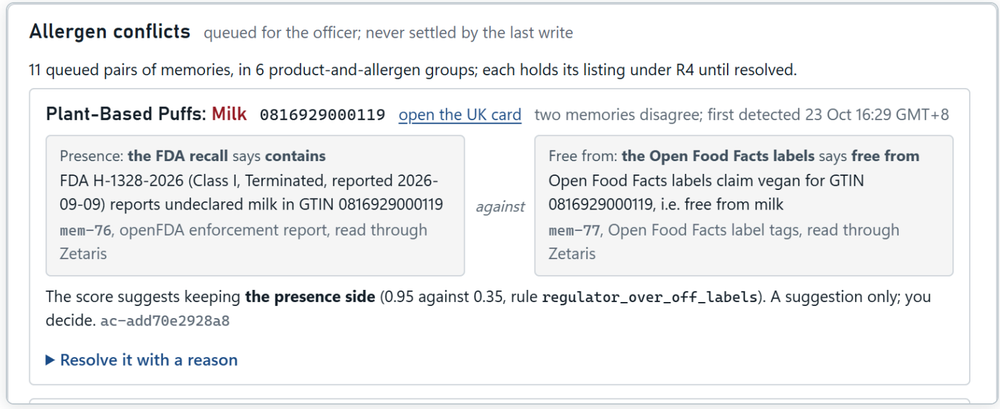
Scout subset: guard and signed contracts

Partly reimplemented from the spec (no reference engine is vendored). The injection guard flags 7/7 Scout injection cases, passes 3/3 clean ones, and flagged 0 of 146 real texts. Contracts are HMAC-signed and single-use.

Prompt tokens per chunk, measured: E4 Lidl biscuits, US

Ollama prompt_eval_count of each chunk's first planner request, and of the same request re-sent with the full history (25 Oct 2026). Dotted: the chars/4 estimate it replaces.





H-MEM conflict queued for the officer: the FDA recall says contains milk; the Open Food Facts labels say vegan. The score only suggests which side to keep.

Conformance: H-MEM 13/13 (vendored and the ClearShelf facade), Markovian 8/8, World Model 8/8, idempotency diff empty. Upstream Meterless/Meterless 1742dec, Apache-2.0.

WHERE NVIDIA CONTRIBUTES

Nemotron 3 Nano 4B runs every model call on the local GPU

159.8 tok/s
extraction median; planner 150.0 to 155.6 median per run
2.0 s median per extraction call

27/27
extraction replies valid JSON on the first try
planner: 57/65 first try, 65/65 after one repair; 25 Oct re-run 59/65, 64/65, 1 fallback step

2,881 MiB
model resident in VRAM, RTX 4070 Ti SUPER 16 GB
Q4_K_M, num_ctx 8192, temperature 0

0
cloud model calls: the client accepts loopback URLs only
one model call at a time

Allergen extraction on 27 SKUs: keyword rule, lexicon alone, Nemotron Q4 and Q8

Measure	K keyword	L lexicon alone	N Q4_K_M	N+V Q4	N Q8_0
"Contains" P / R (80 positives)	0.71 / 0.99	0.97 / 0.97	1.00 / 0.78	1.00 / 0.78	1.00 / 0.84
Trap false alarms (cocoa butter, coconut…)	5/15	1/15	0/15	0/15	0/15
"May contain" read exactly	0/27	26/27	18/27	18/27	16/27
German labels (4), recall	0.95	0.95	0.50	0.50	0.70
English labels (21), recall	1.00	0.98	0.90	0.90	0.90

Two independent readers: together the model and the lexicon missed 1 of the 80 "contains" pairs (the lexicon found 17 of the model's 18 misses, the model 1 of the lexicon's 2), and every disagreement holds the listing. Alone, the lexicon scores best, partly because the reference labels use its rules. Q8_0 lifts German recall from 0.50 to 0.70 at 113 against 154 tok/s and 1,185 MiB more VRAM, so Q4 stays.

Roles

- Multilingual ingredient extraction (thinking off)
- Recall-notice extraction
- Investigation planner (thinking on)
- H-MEM memory mining and one-sentence dream proposals

Cold against warm, same product

Ritz Cracker Sandwiches, US: same 5 chunks and decision; tool-model tokens 3,658 → 804 in and out; wall time 61.1 s → 48.1 s.

Day-0 gate

29/30 replies valid JSON (96.7%) with the adopted settings, Zetaris running on the same PC. The 30B tier was cut: the gate projected too little host RAM.

Thinking, measured per job

Thinking on: 10/10 planner replies valid, but only 2/10 extractions (the reasoning hit the 4,096-token cap). So the planner thinks; extraction does not.

Shared-PC guard

The runner's prompt cache grew host memory by 5,507 MiB over 30 calls. Below 2,048 MiB free, the client unloads the model before the next call.

Model Ollama on 127.0.0.1	Planner measured, median of 5	Peak VRAM measured	Cloud model calls	Cold → warm tokens in+out, measured	Per chunk measured vs est.
nemotron-3-nano:4b	**154 tok/s**	**6,743 MiB GPU, model 2,881 MiB**	**0** 0 non-local refused	chunks **5 → 5** planner **19,348 → 18,576** tool **3,658 → 804** time **61.1 s → 48.1 s**	

The UI's telemetry footer. Peak VRAM is whole-GPU use, including another workload on the shared PC; the model itself holds 2,881 MiB. A full investigation takes 48 to 94 s.

Measured on 23 Oct 2026 (L and Q8_0 columns, language rows and the planner re-run on 25 Oct 2026) with Ollama's token counters and nvidia-smi; reference labels come from recall texts and documented label rules, never typed by hand (docs/eval-report.md, docs/nemotron-bench.md).

FINDINGS FROM LIVE DATA

What the live feeds taught us, each one now handled in the product

1. **Rule against rule: coconut**

   FDA's allergen Q&A (Edition 5, January 2025) removed coconut from the tree-nut list, and UK/EU Annex II does not list it. Recall H-1275-2026 (Lidl's Eridanous apricot biscuits with coconut) still says "tree nut (coconut)". ClearShelf records a rule conflict instead of a tree-nut allergen.

2. **Recalls carry barcodes only some of the time**

   Of the latest 1,000 openFDA "undeclared" reports, 533 mention a UPC, GTIN, EAN or barcode, and 442 contain a code that passes the GS1 check digit.

3. **FSA allergy alerts carry no GTIN**

   Product details hold names only, so an FSA match is name-only: it becomes a review item for the officer (Skinny Crunch, similarity 0.28), never an automatic action.

4. **Crowd tags surface as DQ orphans**

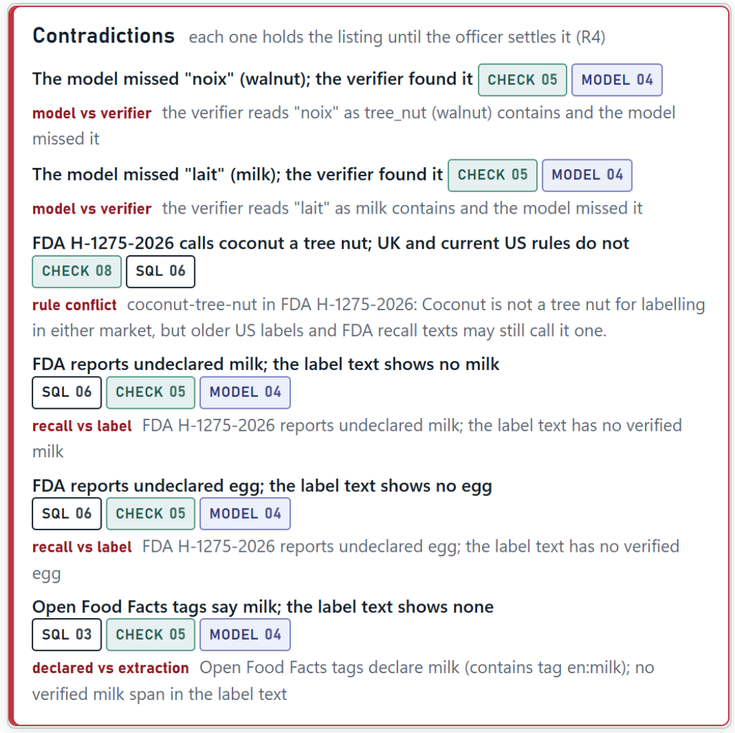
   The Open Food Facts tag `de:blé` on the same biscuits maps to `UK:blé`, which Zetaris DQ reports as a foreign-key orphan, and the listing is held under R3.

5. **Pasted label noise**

   On the same label, an address line pasted into the Open Food Facts ingredient field splits "noix de coco", so the verifier reads "noix" as walnut. The draft marks it "(to confirm)", and a statement with "(to confirm)" cannot be signed.

6. **Sources throttle**

   Open Food Facts search returned HTTP 503, and the product endpoint returned 429 on 10 of 549 curation calls. ClearShelf spaces its calls 5 s apart by default (never below 2.5 s) and caches every table in Zetaris.



Aprikosenkekse mit Kokosraspeln (UK): six contradictions hold the listing, including the coconut rule conflict and two model misses the verifier caught.

1 of 1 row shown

declared_key	gtin	market_code	relation	source_tag
4056489125846|UK|blé|contains	4056489125846	UK:blé	contains	de:blé

SHOW DQ INVALID RECORD on `declared_allergen`: the orphan row, cited in the evidence drawer.

openFDA counts from one 1,000-record page fetched 23 Oct 2026; Open Food Facts call counts from the dev-time curation log.

Decisions you can audit: gated, signed and cited

A data-quality gate on decisions

The clearance USL's DQ rules sit in front of CLEAR: one invalid record holds the listing.

Rule-against-rule conflicts on the record

Regulations that disagree (coconut) are queued like any other contradiction.

Signed, asymmetric autonomy

HOLD and BLOCK run on their own. CLEAR and RELEASE need a single-use HMAC contract bound to the exact values, valid for 15 minutes; a replayed or edited contract is refused.

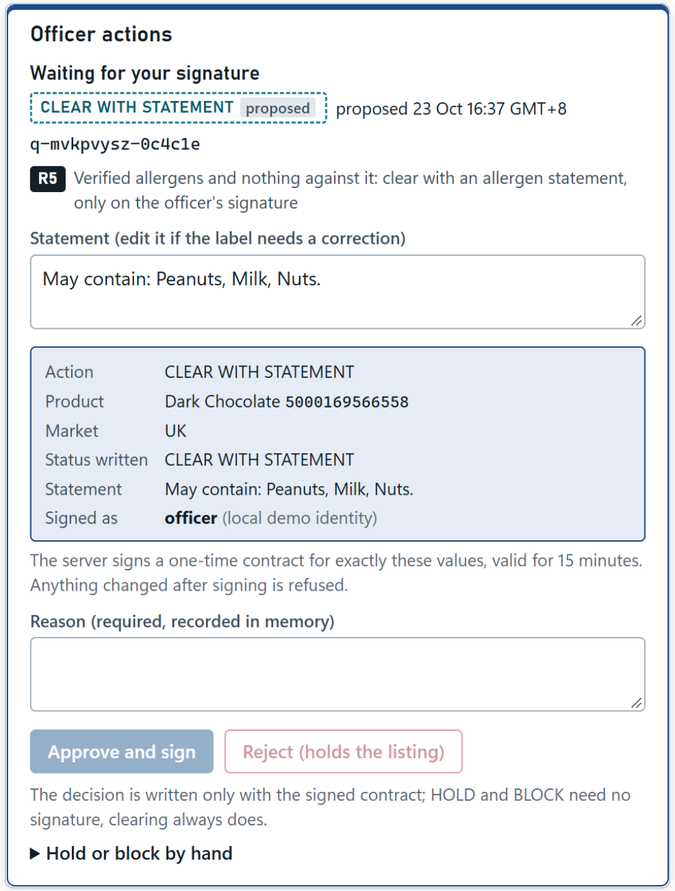
Citations you can re-run

SQL text, sha256 and cache time on every claim; the "why" trace and the ledger point to the same records.

Measured on 23 Oct 2026

Scenarios E1–E10	**10/10 pass**
Card claims cited (E-cov)	100%
Policy firings / review items cited	43/43, 37/37
Records with every required field	200/200
Cited SQL re-run returns the rows (E-replay)	95/95
Contradictions surfaced (E-contra)	3/3
Spans not in the source, kept	0 (1 dropped, counted)
CLEAR on a product with an ongoing exact recall	**0**
CLEAR or RELEASE executed without a signed contract	**0**
Zetaris `signed_clear`: signed row / synthetic unsigned row	passed / flagged
Agent tests / gateway tests	419 / 199 pass (4 skipped)

E8 is a synthetic injection fixture, labelled and eval-only. E2 is a replay and is labelled as one on every row it writes.



What the officer signs: action, product, market, status and statement. A reason is required and recorded in memory.

Scenario run of 23 Oct 2026, 17:06–17:16 SGT, re-run on 25 Oct 2026 on commit aeb5b92 with the same counts; signed_clear checks 20:38–20:40 SGT; docs/eval-report.md sections 1, 2 and 7.

LIMITS AND FUTURE ENHANCEMENTS

What it does not do yet, and what comes next

Limits, measured

Recall is the weak side

Nemotron's "contains" recall is 0.78 overall and 0.58 on the six non-English labels; the lexicon verifier alone reached 0.97 (partly circular). Misses are held for the officer, never cleared.

About a minute per product

Investigations take 48 to 94 s on the local 4B model, one call at a time.

A small evaluation set

Extraction is scored on 27 products; the 10 eval-only products are registered in Zetaris but not yet in the USL product table.

Demo identity and admin reads

The officer is a local demo identity with no sign-on, and Zetaris reads run on the admin login behind a read-only allowlist.

Crowd-sourced label data

Open Food Facts can lag the physical label, and lot codes are not matched. Every card lists what was not checked.

Decision support only, not legal advice. ClearShelf never publishes a free-from claim on its own.

Future enhancements

Supplier PIM and spec-sheet connector

Replace the Open Food Facts stand-in with the grocer's own supplier feed, registered in Zetaris the same way.

Lot and batch-code matching

Match recall code information against stock, so a recall holds only the affected lots.

EU RASFF and more markets

Add RASFF notifications as a third notice feed, and further statutory allergen lists to the reference table.

Production identity and least privilege

Single sign-on in front of the officer endpoints, and a least-privilege Zetaris reader role.

Better non-English recall

Adopt the Q8 build (on 27 labels, recall 0.84 against 0.78 for Q4 and 0.70 against 0.50 on German, at 113 against 154 tok/s) or a larger tier, and measure the thinking ablation on multilingual labels.

Also not measured yet: cold against warm on a fresh memory store, and officer time saved (docs/eval-report.md, "Not measured").

Built in a Cursor workspace; every source attributed

ClearShelf was developed in a Cursor workspace with project rules in `.cursor/rules` and `AGENTS.md`. Cursor Agent ran the Zetaris JDBC preflight against our local instance. It drafted the decision write-back SQL, the per-product Open Food Facts registration template and the DQ rule documentation, with static tests for the SQL. It also drafted the multilingual allergen lexicon, the jurisdiction rules, the GTIN normaliser and the claim tester, with their tests. Its Free plan then reached its usage limit, so the remaining planned Cursor tasks were not run by Cursor Agent. After an upgrade to the Pro plan, Cursor Agent ran C9 on 25 Oct 2026: clock-independent runtime tests. Every Cursor task, its commit and what was corrected afterwards are listed in `docs/how-cursor-was-used.md`.

Cursor Agent run	Scope
G9	Task index in docs/cursor-tasks/README.md
C1	Zetaris JDBC preflight against our local instance, read-only
C2	Decision write-back SQL, Open Food Facts registration template, DQ rule documentation, static SQL tests
C3	Multilingual allergen lexicon, jurisdiction rules, GTIN normaliser, claim tester, with tests
C9	Clock-independent runtime tests (after the submission, Pro plan)

Project rules in `.cursor/rules`

`zetaris.mdc` Lightning SQL rules · `meterless.mdc` engine contracts and the never-edit rule for vendored engines · `safety.mdc` decision safety and risk asymmetry · `data-licences.mdc` attribution and what data may be committed · `style.mdc` code, test and commit conventions

Credits and licences

Open Food Facts: data from openfoodfacts.org, ODbL 1.0. No bulk data in the repository.

openFDA food enforcement reports: public domain / CC0 1.0. ClearShelf is not endorsed by the FDA.

UK FSA food alerts: contains public sector information licensed under the Open Government Licence v3.0.

Meterless H-MEM, Markovian and World Model reference engines: Apache-2.0, vendored unchanged.

Zetaris getting-started SQL helper, adapted: Apache-2.0.

NVIDIA Nemotron 3 Nano: NVIDIA Open Model License; weights not redistributed.

ClearShelf: Apache-2.0.

Umar Mohktar, Singapore - solo